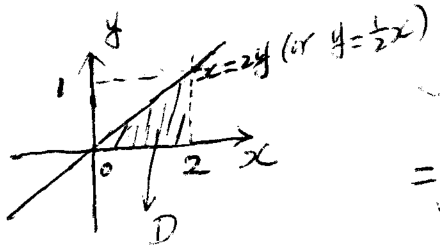
Math 550, S20 **Name:** Answer

Quiz 7

Instructions: **There are totally 3 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (3 points) Given an double integral $\int_0^1 \int_{2y}^2 \cos(x^2)\, dxdy$. Reverse the order of the integration and evaluate the resulted iterated integral.



$$\int_0^1 \int_{2y}^2 \cos(x^2)\,dx\,dy$$

$$= \int_0^2 \int_0^{\frac{1}{2}x} \cos(x^2)\,dy\,dx = \int_0^2 \cos(x^2)\,[y]_{y=0}^{y=\frac{1}{2}x}\,dx$$

$$= \int_0^2 \frac{1}{2}x \cos x^2\,dx$$

$$= \left[\frac{\sin x^2}{4}\right]_{x=0}^{x=2}$$

$$= \frac{\sin 4}{4}$$

Problem 2. (3 points) Evaluate the iterated integral $\int_0^1 \int_0^z \int_0^y 2ze^{-y^2}\, dxdydz$.

$$\int_0^1 \int_0^z \int_0^y 2ze^{-y^2}\,dx\,dy\,dz = \int_0^1 \int_0^z 2ze^{-y^2}[x]_{x=0}^{x=y}\,dy\,dz$$

$$= \int_0^1 \int_0^z 2zye^{-y^2}\,dy\,dz$$

$$= \int_0^1 z(-e^{-y^2})\Big|_{y=0}^{y=z}\,dz$$

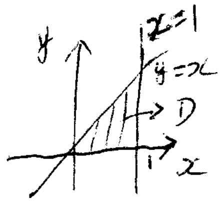
$$= \int_0^1 -z(e^{-z^2}-1)\,dz$$

$$= \int_0^1 -ze^{-z^2}+z\,dz = \left[\frac{1}{2}e^{-z^2}+\frac{z^2}{2}\right]_{z=0}^{z=1}$$

$$= \left(\frac{e^{-1}}{2}+\frac{1}{2}\right)-\left(\frac{1}{2}+0\right) = \frac{1}{2e}$$

Problem 3. (4 points) Evaluate the triple integral $\iiint_W 6xy\, dV$ where W the region which lies under the plane $z = 1 + x + y$ and above the region in the xy-plane enclosed by the curves $y = x$, $y = 0$, and $x = 1$.

$$D = \{(x,y) \mid 0 \le x \le 1,\ 0 \le y \le x\}$$



$$W = \{(x,y,z) \mid (x,y) \in D,\ 0 \le z \le 1+x+y\}.$$

Thus

$$\iiint_W 6xy\,dV = \iint_D \left[\int_0^{1+x+y} 6xy\,dz\right] dA$$

$$= \int_0^1 \int_0^x \int_0^{1+x+y} 6xy\,dz\,dz\,dy\,dx.$$

$$= \int_0^1 \int_0^x 6xy\,[z]_{z=0}^{z=1+x+y}\,dy\,dx$$

$$= \int_0^1 \int_0^x 6xy + 6x^2y + 6xy^2\,dy\,dx$$

$$= \int_0^1 \left[3xy^2 + 3x^2y^2 + 2xy^3\right]_{y=0}^{y=x} dx$$

$$= \int_0^1 3x^3 + 5x^4\,dx$$

$$= \left[\frac{3}{4}x^4 + x^5\right]_{x=0}^{x=1}$$

$$= \left(\frac{3}{4} + 1\right) - 0 = \frac{7}{4}$$